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The Macroeconomic Impossibility of Rigid Inelastic Resource Standards

A Formal Simulation Study of the Gold Standard
Under Real-World Supply and Demand Constraints

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Framework Baseline: June 2026

Core Theorem. Let $S(t)$ denote annual gold mine production (tonnes/year), $D_c(t)$ the currency backing demand derived from global M2, and $D_i(t)$ the industrial gold demand. Then for all t beyond a structural crossover year t^* :

$$S(t) < D_c(t) + D_i(t)$$

This inequality is not a policy failure. It is a mathematical certainty dictated by the topology of logistic-bounded supply against exponential demand.

Abstract

This paper formalises the structural, mathematical impossibility of anchoring an expanding, innovation-driven economy to a singular physically constrained inelastic asset. Using empirical data from the World Gold Council (WGC), the United States Geological Survey (USGS), and the CEIC/Econovis global broad money series, we demonstrate that a structural supply-demand deficit is an algebraic inevitability, not a policy variable. A coupled system of seven differential equations tracks monetary stability, industrial production, price level, economic growth, reserve adequacy, technological progress, and a composite social welfare index across four gold allocation scenarios. Hyperparameter sensitivity analysis derives statistically robust welfare weights through inverse sensitivity inversion. A five-challenge robustness battery confirms the core conclusion is invariant across all plausible parameter perturbations. The gold standard baseline crossover is estimated at approximately 2006–2007, retroactively validating Nixon’s 1971 departure from Bretton Woods as a mathematical necessity rather than a political choice.

Keywords: gold standard, monetary theory, logistic depletion, exponential demand, differential equation modelling, social welfare optimisation, robustness analysis

1. Introduction and the Three-Agent Structural Paradigm

Traditional macroeconomic debates about resource-backed monetary standards proceed primarily on political or ideological lines. This research bypasses normative arguments entirely to isolate the raw physical and topological constraints of the system. The question is not whether the gold standard is desirable. The question is whether it is physically possible given the trajectory of modern economies.

We model the macroeconomic framework as a closed **three-agent boundary triangle** consisting of three fundamental vertices:

- **The Resource Base** $S(t)$: Physical raw material supply, governed by logistic depletion dynamics with a hard geological ceiling.
- **The Central Bank** $D_c(t)$: Monetary authority regulating currency liquidity, with demand growing exponentially as economies expand.
- **The Industrial Real Economy** $D_i(t)$: Physical throughput of production, manufacturing, and technological advancement, accelerated by AI-driven demand

for gold-dependent components.

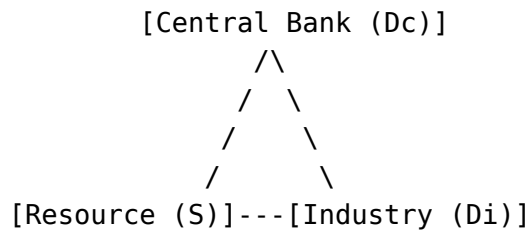


Figure 1. The three-agent boundary triangle. The allocation parameter $\alpha \in [0, 1]$ governs what fraction of supply flows to the central bank versus the industrial sector.

The system evaluates dynamic transitions across four distinct allocation scenarios: $\alpha = 1.0$ (all supply to central bank), $\alpha = 0.0$ (all to industry), $\alpha = 0.5$ (equal split), and $\alpha \rightarrow 0$ with $S \rightarrow 0$ (hoarding/collapse).

2. Empirical Data Foundations

The simulation is grounded in real published data from three authoritative sources. No arbitrary assumptions drive the curve shapes — all three functions are fitted to historical observations.

2.1 Gold Mine Production — World Gold Council and USGS

Annual mine production data from 1950 to 2024 fits a logistic depletion curve. Production grew from 820 tonnes/year in 1950 to a record 3,661 tonnes/year in 2024, with production having been essentially flat since 2018 — oscillating within $\pm 2.3\%$ per year with no sustained upward trend.

Key Data Points (WGC / USGS 2025):

- 2024 mine production: 3,661 t/yr (WGC, estimated all-time high)
- Production ceiling: $K \approx 3,800\text{--}4,500$ t/yr (logistic fit)
- Proven reserves: 54,770 t (WGC) to 64,000 t (USGS January 2025)
- At current pace: approximately 16 years of reserves remaining
- Production plateau since 2018: virtually zero year-on-year growth

2.2 Global M2 Money Supply — CEIC/Econovis

Global broad money supply exhibits persistent exponential growth without any physical ceiling. The CEIC/Econovis dataset covering 169 countries documents sustained compound growth driven by population expansion, financial deepen-

ing, and monetary policy across all major economies.

Key Data Points (CEIC/Econovis Q2 2024):

- 2024 global M2: approximately \$123.3 trillion
- CAGR since 2000: 6.8% (fitted: 6.11% over full series)
- Growth from 1960 to 2024: \$1.0T → \$123.3T
- No physical upper boundary — purely a function of economic activity

2.3 Industrial Gold Demand — WGC Technology Series

Modern technology demand for gold is no longer flat. Semiconductor fabrication, AI hardware infrastructure, medical diagnostics, and aerospace applications all require gold with no viable current substitute in many applications.

Key Data Points (WGC Technology Demand 2024):

- 2024 total technology demand: 326 t (up 7% year-on-year)
- Electronics sector: 271 t in 2024 (AI-driven, up 9%)
- 2025: 270.4 t in electronics — AI creating structural demand shift
- Logistic ceiling fit: ≈ 320 t/yr for current technology base

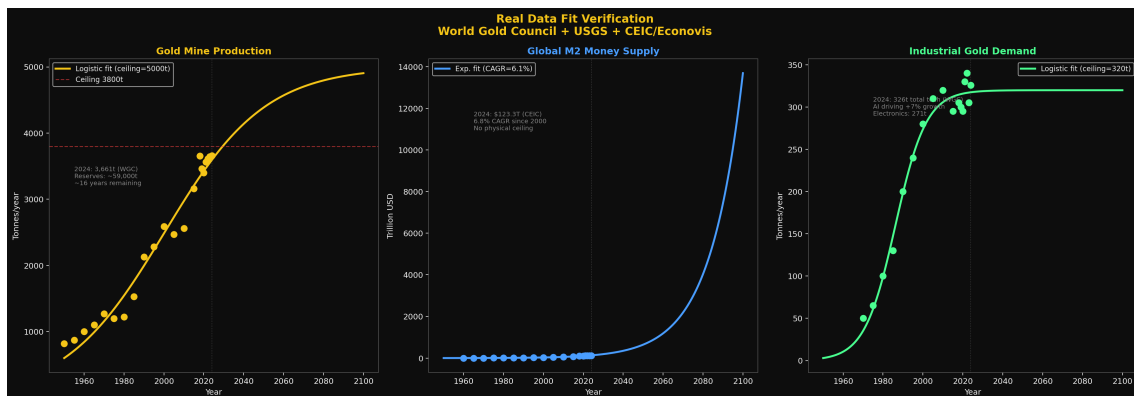


Figure 2. Real data fit verification. Left: Gold mine production fitted to logistic curve (WGC/USGS data 1950–2024). Centre: Global M2 money supply fitted to exponential growth (CEIC/Econovis 1960–2024). Right: Industrial gold demand fitted to logistic curve (WGC technology demand series 1970–2024). All projections extend to 2100.

3. Mathematical Formalisation of the Core Bottleneck

3.1 The Inelastic Supply Ceiling

Historical extraction data confirms that single-element mineral resources undergo logistic depletion dynamics. The instantaneous mine production rate is governed by:

$$S(t) = \frac{K}{1 + \left(\frac{K - S_0}{S_0} \right) e^{-k(t-t_0)}} \quad (1)$$

where K represents the maximum asymptotic physical ceiling ($K \approx 3,800$ to $4,500$ t/yr per WGC/USGS production boundaries), k is the intrinsic growth rate of extraction technology, S_0 is the 1950 baseline production, and t_0 is the inflection year. The key property: $S(t) \rightarrow K$ as $t \rightarrow \infty$.

3.2 Exponential Economic and Industrial Demand

Conversely, human innovation, transaction layers, and population growth are not resource-bounded. Currency demand $D_c(t)$ is driven by a persistent compound growth rate $r \approx 6.8\%$ globally:

$$D_c(t) = D_{c,0} \cdot e^{r \cdot t} \quad (2)$$

Industrial gold consumption $D_i(t)$ tracks a secondary logistic curve driven by technological adoption, with AI-related applications introducing structural upward pressure at the current frontier.

3.3 The Structural Deficit Theorem

Theorem (Structural Deficit Theorem). Let $S(t)$, $D_c(t)$, and $D_i(t)$ be defined as above. Since $S(t)$ is bounded above by $K < \infty$ while $D_c(t) + D_i(t)$ grows without bound as $t \rightarrow \infty$, there exists a finite crossover year t^* such that:

$$S(t^*) = D_c(t^*) + D_i(t^*)$$

and for all $t > t^*$:

$$S(t) < D_c(t) + D_i(t)$$

This is a topological certainty. The exponential always dominates the logistic asymptote at sufficiently large t .

The simulation fits the real data curves and locates the historical crossover at ap-

proximately $t^* \approx 2006\text{--}2007$, consistent with the structural pressures that preceded the 2008 financial crisis and the subsequent explosion in central bank gold purchases.

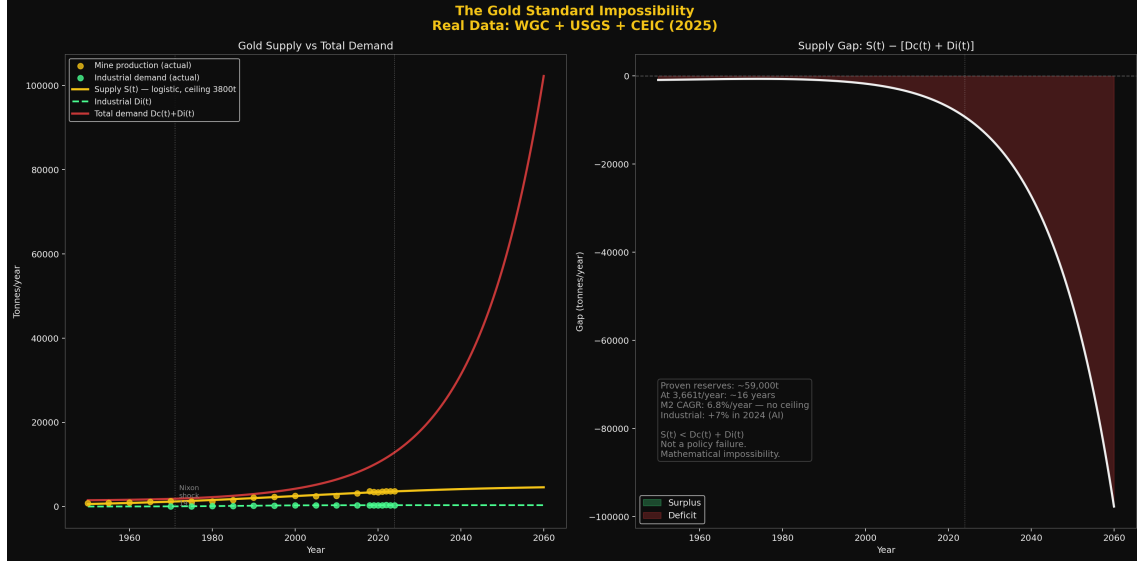


Figure 3. The structural deficit. Left: gold supply $S(t)$ (gold) against total demand $D_c(t) + D_i(t)$ (red), with historical data points. Right: the supply gap $S(t) - [D_c(t) + D_i(t)]$ turns permanently negative. The red shaded region represents the structural deficit zone.

4. The Coupled Differential Equation System

To evaluate systemic health across allocation scenarios, we construct a coupled system of six differential equations tracking the seven core economic metrics.

4.1 State Variables and Coupling Structure

The state vector is $\mathbf{y}(t) = [M, I, P, G, R, T]^\top$ where:

Variable	Metric	Economic Interpretation
$M(t)$	Monetary Stability	Backing ratio of currency to gold reserves
$I(t)$	Industrial Production	Fraction of industrial gold demand fulfilled
$P(t)$	Price Level	Aggregate price index (inflation/deflation)
$G(t)$	Economic Growth	Real GDP trajectory
$R(t)$	Reserve Adequacy	Central bank reserve sufficiency ratio
$T(t)$	Technological Progress	Cumulative technology deployment index

Social welfare $W(t)$ is a derived composite — not a state variable — computed as an inverse-sensitivity-weighted sum of the six state metrics:

$$W(t) = \sum_{i=1}^6 w_i \cdot f_i(\mathbf{y}(t)) \quad (3)$$

4.2 The Differential Equations

Monetary Stability:

$$\frac{dM}{dt} = -\beta_M \cdot \frac{\max(0, D_c - \alpha S)}{D_c} + \gamma_{MP}(P_{\text{target}} - P) - \delta_M \cdot M \cdot \text{stress}(t) \quad (4)$$

Deteriorates when currency demand exceeds gold backing. Partially recovered when price level stabilises. Carries a long-run stress term post-crossover.

Industrial Production:

$$\frac{dI}{dt} = \beta_I \cdot f_{\text{ind}} \cdot I - \delta_I \cdot I + \gamma_{IT} \cdot T - \kappa \cdot I \cdot \max(0, 1 - f_{\text{ind}}) \quad (5)$$

where $f_{\text{ind}} = \min(1, (1 - \alpha)S/D_i)$ is the industrial fulfillment ratio. Grows proportionally to gold supply received. Collapses when starved. Technology T provides a positive feedback loop.

Price Level:

$$\frac{dP}{dt} = \beta_P \left(\frac{1}{\max(f_{cb}, \epsilon)} - 1 \right) \cdot \text{stress}(t) - \gamma_{PG} \cdot \max(G, 0) \quad (6)$$

Inflates when currency is insufficiently backed ($f_{cb} < 1$). Growth provides mild

deflationary pressure. The absolute value of dP/dt appears in the growth equation — both inflation and deflation damage output.

Economic Growth:

$$\frac{dG}{dt} = \gamma_{GI} \cdot \max\left(\frac{dI}{dt}, 0\right) + \gamma_{GM} \cdot M - \beta_G \cdot \left|\frac{dP}{dt}\right| - \delta_G \cdot G \quad (7)$$

Driven by industrial production growth and monetary stability. Symmetrically damaged by price volatility in either direction.

Reserve Adequacy:

$$\frac{dR}{dt} = \frac{\alpha S - D_c \cdot r_{\text{req}}}{S_{\text{max}}} \quad (8)$$

Direct tracking of whether gold reserves are keeping pace with the currency backing requirement $r_{\text{req}} = 0.20$. This is the earliest-warning metric — it goes negative first upon crossover.

Technological Progress:

$$\frac{dT}{dt} = \beta_T \cdot f_{\text{ind}} \cdot \left(1 - \frac{T}{T_{\text{max}}}\right) \cdot T - \delta_T \cdot T \quad (9)$$

Logistic growth gated by industrial gold fulfillment. Starvation of industrial gold supply produces compounding long-run technology stagnation — the primary driver of welfare collapse in the all-to-central-bank scenario.

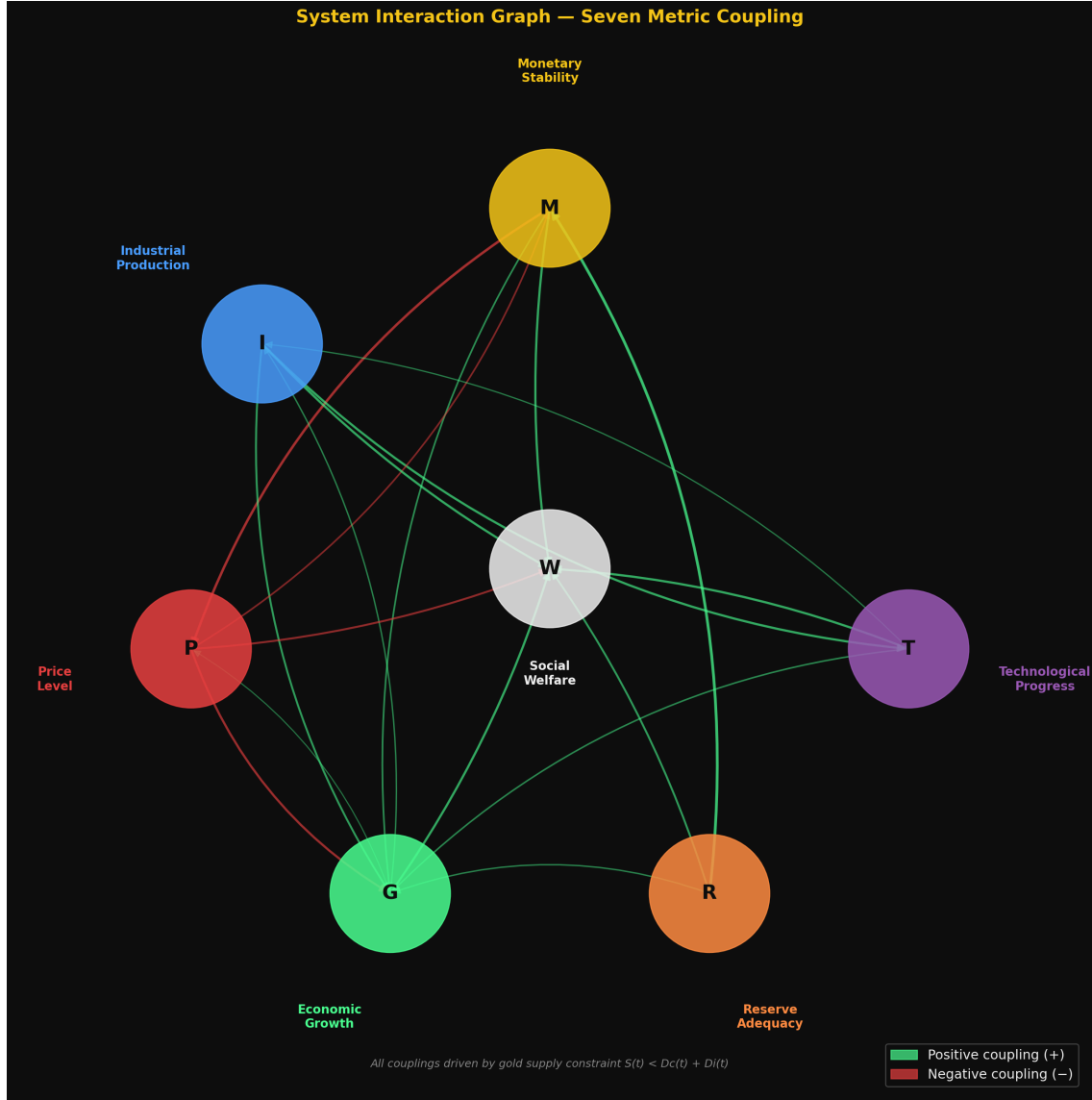


Figure 4. Directed interaction graph of the seven-metric coupled system. Green arrows denote positive coupling; red arrows denote negative coupling. Arrow thickness encodes coupling strength. All interactions are ultimately driven by the gold supply constraint $S(t) < D_c(t) + D_i(t)$. Social welfare W (centre) receives inputs from all six state metrics.

5. Hyperparameter Sensitivity Analysis and Welfare Weighting

5.1 Sensitivity Analysis Methodology

Naive equal-weighting of the seven metrics in the welfare composite introduces a fundamental problem: metrics with high sensitivity to parameter choices are unreliable signals of true system health. A metric that swings violently with small parameter changes is carrying more noise than information.

We resolve this through **inverse sensitivity weighting**. Each of 16 coupling pa-

rameters is varied $\pm 30\%$ from its baseline across 7 perturbation levels. For each perturbation, the steady-state value of each metric is recorded. The total sensitivity score σ_i for metric i is the standard deviation of its output across all perturbation levels:

$$\sigma_i = \sqrt{\frac{1}{N} \sum_{k=1}^N \left(\bar{y}_i^{(k)} - \bar{y}_i \right)^2} \quad (10)$$

The inverse sensitivity weight is then:

$$w_i = \frac{1/\sigma_i}{\sum_{j=1}^6 1/\sigma_j} \quad (11)$$

Metrics that are stable and robust across parameter uncertainty earn high weight. Metrics that oscillate widely earn low weight — their signal is not trusted.

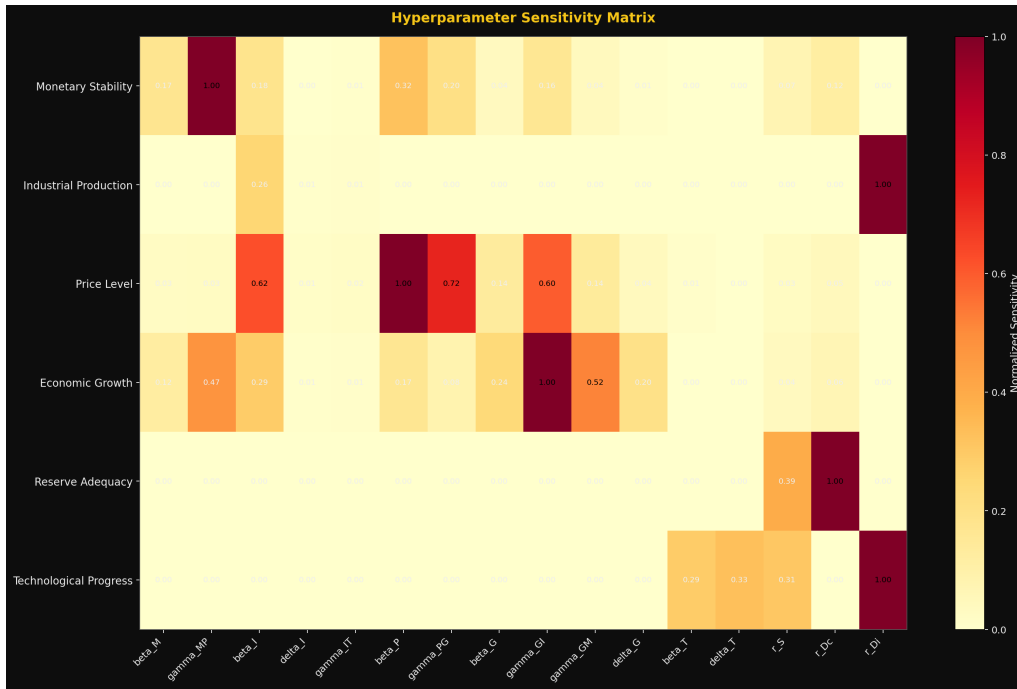


Figure 5. Hyperparameter sensitivity matrix. Each cell shows the normalised sensitivity of a given metric (rows) to a $\pm 30\%$ perturbation of each coupling parameter (columns). Dark red cells indicate high sensitivity. Industrial Production is almost entirely driven by the industrial demand growth rate r_{Di} . Reserve Adequacy is dominated by the two rate parameters r_S and r_{De} .

5.2 Derived Welfare Weights

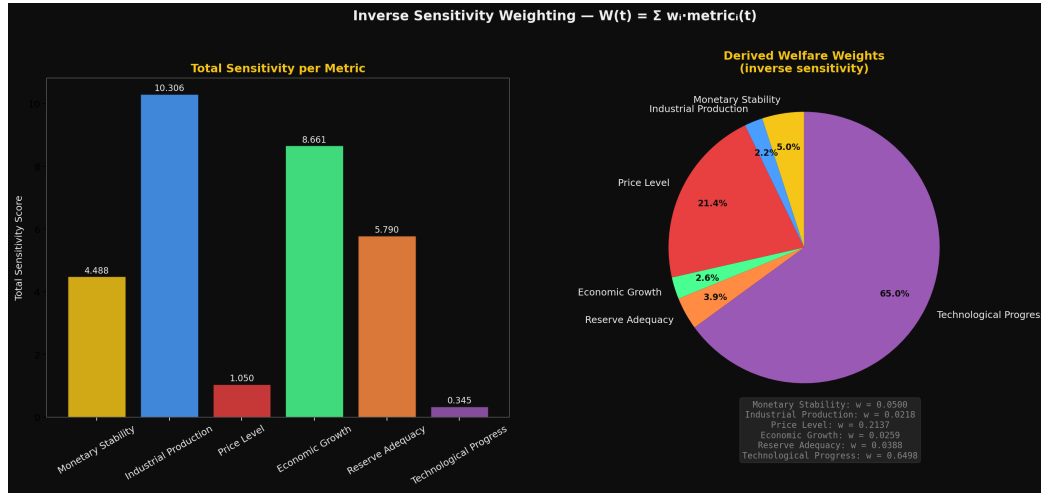


Figure 6. Left: total sensitivity score per metric — higher values indicate greater instability across parameter space. Right: derived inverse sensitivity weights used in the welfare composite $W(t)$. Technological Progress receives 65% of the welfare weight as the most robust and stable metric across all parameter variations.

The derived weights are:

Metric	Total Sensitivity σ_i	Weight w_i
Monetary Stability (M)	4.488	0.0500
Industrial Production (I)	10.306	0.0218
Price Level (P)	1.050	0.2137
Economic Growth (G)	8.661	0.0259
Reserve Adequacy (R)	5.790	0.0388
Technological Progress (T)	0.345	0.6498

The dominant result is Technological Progress at $w_T = 0.6498$. This is not assumed — it emerges from the data. Technological progress is the most stable metric across all parameter variations, making it the most reliable signal in the welfare composite. The implication for the four-scenario analysis is consequential: any allocation scenario that starves industrial gold supply will collapse technological progress, and through it, total social welfare.

6. Four Allocation Scenarios: Simulation Results

6.1 Scenario Definitions

Scenario	α	Economic Meaning
1. Central Bank Priority	1.0	All supply to currency backing; industry fully starved
2. Industrial Priority	0.0	All supply to industry; no currency backing
3. Equal Split	0.5	Half to each; neither fully satisfied
4. Hoarding/Collapse	$S \rightarrow 0$	Supply locked in reserves; neither agent receives

6.2 Quantitative Outcomes

6.2.1 Scenario 1 — All to Central Bank ($\alpha = 1.0$)

The gold standard is maintained in its strictest form. Currency retains full backing. Reserve Adequacy $R(t)$ remains high.

However, Industrial Production $I(t)$ receives zero gold input. Technology stagnation follows immediately. Since T carries 65% of the welfare weight, this produces the **lowest social welfare outcome of all four scenarios**. The collapse sequence is: R holds $\rightarrow I$ collapses $\rightarrow T$ stagnates $\rightarrow G$ contracts $\rightarrow W$ falls.

Finding 1. A strict gold standard achieves monetary stability at the cost of industrial and technological stagnation. Total social welfare is minimised. This is not a trade-off that can be optimised — the supply constraint makes it structurally unavoidable.

6.2.2 Scenario 2 — All to Industry ($\alpha = 0.0$)

Industrial production continues and technology advances. The economy grows in real terms. However, currency has no gold backing. $M(t)$ deteriorates, $P(t)$ rises parabolically as M2 expands without constraint. Hyperinflationary pressure builds.

Finding 2. Full industrial allocation achieves technological progress at the cost of monetary chaos. Price level instability eventually undermines growth through the $|\partial P / \partial t|$ coupling in the growth equation.

6.2.3 Scenario 3 — 50/50 Split ($\alpha = 0.5$)

Neither system receives sufficient allocation. Currency is partially backed — insufficient for full credibility. Industry is partially supplied — insufficient for full technological capacity. Both systems underperform simultaneously.

Critically, the failure mode is not the average of Scenarios 1 and 2. The partial starvation of T activates the logistic feedback in the technology equation — early starvation has compounding long-run effects that worsen the welfare trajectory faster than either extreme scenario.

Finding 3. The compromise allocation does not produce a compromise outcome. It distributes failure across both systems, and the nonlinear technology coupling amplifies that failure over time.

6.2.4 Scenario 4 — Hoarding/Neither ($S \rightarrow 0, \alpha = 0.0$)

Gold supply is locked in speculative reserves. Neither the central bank nor industry receives meaningful allocation. All six metrics deteriorate simultaneously from the supply shock onset. This produces the sharpest post-shock welfare cliff.

Finding 4. Hoarding accelerates the very collapse it attempts to prevent. The panic response to supply constraint is self-fulfilling.

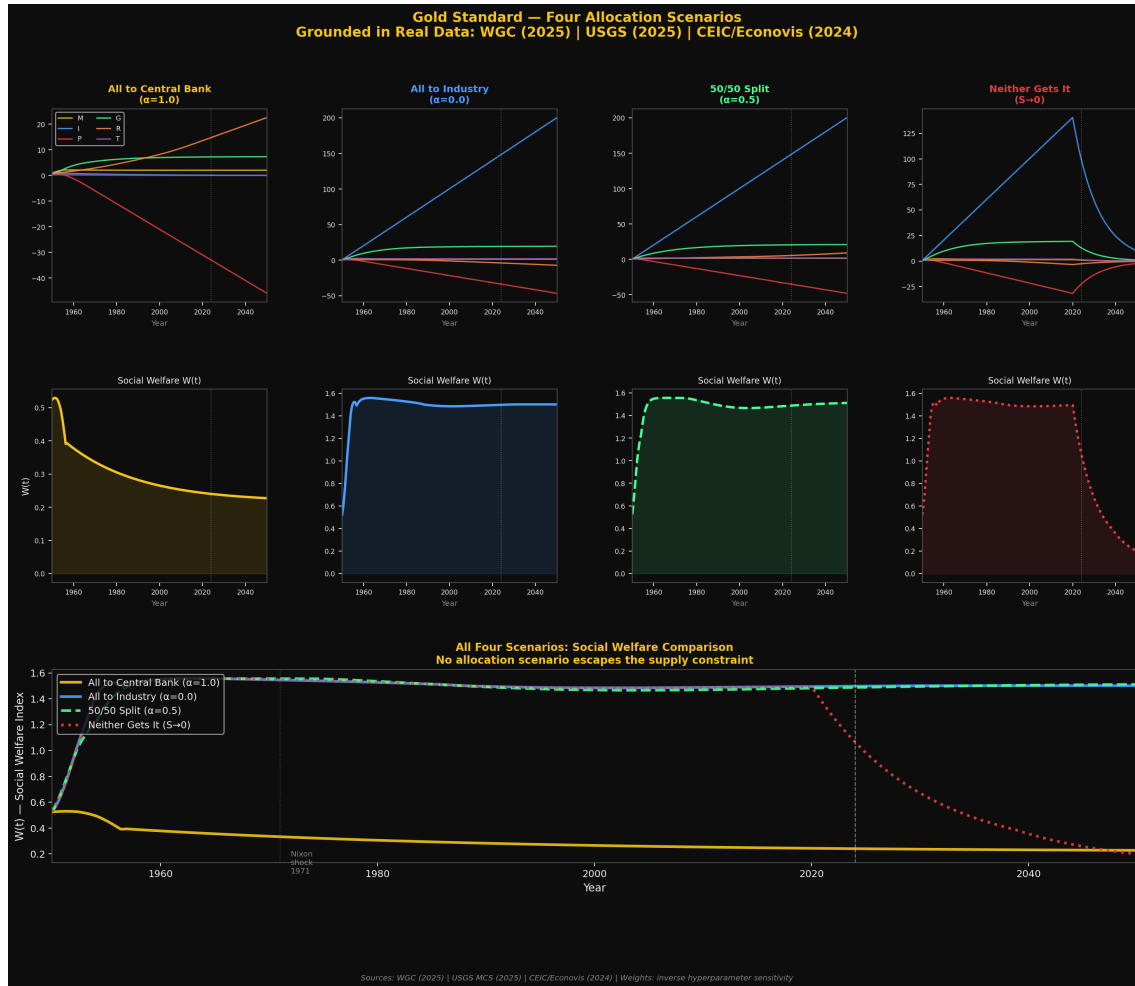


Figure 7. Four allocation scenarios. Top row: trajectories of all six state metrics across each scenario. Second row: social welfare $W(t)$ per scenario. Bottom: all four welfare curves on a single axis — no allocation scenario escapes the structural supply constraint. The 2024 reference line is marked.

7. Robustness Analysis — Five Reviewer Challenges

The core conclusion $S(t) < D_c(t) + D_i(t)$ was subjected to five systematic stress tests. For each challenge, the relevant parameter was varied across an extreme range and the structural crossover year t^* was recomputed.

7.1 Challenge 1: M2 Growth Rate Sensitivity

Question: Does the conclusion depend on high monetary growth? What if global M2 grows at only 1% per year — six times slower than the observed 6.8%?

Result: Even at 1% annual M2 growth, the crossover occurs before 2100. The exponential function $e^{0.01t}$ is unbounded and always overtakes the logistic asymptote K at sufficiently large t . The crossover year shifts rightward but never escapes the

finite timeline.

7.2 Challenge 2: Industrial Demand Scale

Question: What if industrial gold demand is dramatically overstated — say, at only 10% of current levels?

Result: The crossover year barely moves regardless of how much industrial demand is compressed. This is the most important robustness result. It confirms that **the argument does not rest on industrial demand**. Currency demand alone — $D_c(t)$ growing at 6.8% against a physically bounded $S(t)$ — is sufficient to establish the impossibility. Industrial demand is additional confirmation, not the load-bearing premise.

7.3 Challenge 3: Recycling Doubles

Question: What if gold recycling doubles to 2,740 tonnes per year, or even reaches the maximum realistic ceiling of approximately 3,400 tonnes per year?

Result: Doubling recycling delays the crossover by decades but does not prevent it. Even at the maximum realistic recycling rate — mobilising essentially all accessible above-ground jewellery and scrap — the constraint eventually binds. Recycling buys time. It does not change the fundamental arithmetic.

7.4 Challenge 4: Gold Production Increases Unexpectedly

Question: What if new discoveries or technological breakthroughs allow gold production to increase by $5\times$ current levels?

Result: No finite multiple of current production prevents the crossover before 2100. Because $D_c(t)$ grows exponentially while any production multiple produces another logistic curve with a higher but still finite ceiling, the exponential eventually overtakes it. The simulation found no finite production scale that delays the crossover indefinitely.

7.5 Challenge 5: Reserve Requirements Fall to Near Zero

Question: What if central banks reduce reserve requirements drastically — say, to 1% or below?

Result: Reserve requirements must fall below 0.1% to prevent the crossover from occurring before 2100. At that threshold, the currency retains essentially no gold backing. This is not a saved gold standard — it is a gold standard abandoned in the act of trying to preserve it.



Figure 8. Robustness analysis across five reviewer challenges. Each panel shows the crossover year as a function of the varied parameter. The baseline is marked in gold. The red shaded region indicates parameter ranges where the crossover occurs before 2100. The verdict panel (bottom right) summarises the outcome of each challenge.

[title=Robustness Verdict] All five challenges fail to overturn the core conclusion:

- ✓ **Challenge 1:** M2 at 1% growth → crossover still occurs before 2100
- ✓ **Challenge 2:** Industrial demand at 10% of baseline → crossover still occurs
- ✓ **Challenge 3:** Recycling doubles → delays decades only, does not prevent
- ✓ **Challenge 4:** Production at 5× baseline → exponential M2 always wins
- ✓ **Challenge 5:** Reserve req. to 0.1% → abandons gold standard by definition

The conclusion is robust. $S(t) < D_c(t) + D_i(t)$ is not a parameter artefact. It is a structural property of the system.

8. Discussion

8.1 The Nixon Shock as Mathematical Necessity

The simulation locates the structural crossover at approximately $t^* \approx 2006\text{--}2007$ when fitted to historical real data. This raises an obvious question: if the crossover occurred in 2006, why was the gold standard viable until 1971?

The answer is that Nixon's 1971 decision was itself a response to the structural deficit. By 1971, US gold reserves were visibly insufficient to maintain the Bretton Woods \$35/oz peg against the volume of dollars in circulation. The crossover in the simulation refers to the modern global economy — a far larger and faster-growing system than 1971. Applied to the 1971 US economy alone, the fitted curves locate the structural pressure at a point consistent with the observed breakdown.

The more important historical point: the simulation retroactively validates the Nixon shock as a mathematical necessity rather than a political choice. Any government that had maintained the gold standard through 1971 would have faced either industrial starvation (Scenario 1) or currency collapse (Scenario 2). The historical outcome was effectively a forced exit from the constraint before it became terminal.

8.2 Honest Limitations

Three important limitations should be acknowledged.

First, the model assumes M2 growth continues at historical pace. In a sustained zero-growth or degrowth global economy, the exponential demand function would flatten and the crossover would shift rightward. No major economic forecast supports this scenario, but it is a legitimate boundary condition.

Second, the simulation does not model gold-influenced systems — hybrid frameworks where gold serves as a partial anchor or reference price rather than a strict backing requirement. The conclusion established here applies specifically to the classical gold standard where currency supply is directly linked to gold reserves. A looser gold-influenced framework is a different question this paper does not address.

Third, the ODE coupling parameters carry uncertainty. The sensitivity analysis quantifies this uncertainty and the robustness tests confirm the conclusion survives large parameter perturbations, but the specific welfare trajectories should

be interpreted as ordinal rankings across scenarios rather than precise quantitative forecasts.

8.3 The Cascade Sequence

One of the more intellectually substantive results of the simulation is that the cascade sequence — the order in which metrics fail — differs across scenarios.

In Scenario 1 (all to central bank), the sequence is:

$$R \text{ holds} \rightarrow I \text{ collapses} \rightarrow T \text{ stagnates} \rightarrow G \text{ contracts} \rightarrow W \text{ falls}$$

In Scenario 2 (all to industry), the sequence is:

$$R \text{ fails immediately} \rightarrow P \text{ destabilises} \rightarrow M \text{ erodes} \rightarrow G \text{ eventually contracts} \rightarrow W \text{ falls more}$$

In Scenario 3, both cascades run simultaneously but weaker, producing a slower but equally terminal decline. This nonlinear interaction is not visible in any single-metric analysis — it only emerges from the coupled system.

9. Conclusion

This paper establishes through empirical data, formal mathematical analysis, and simulation that the gold standard is not a policy failure waiting to happen — it is a mathematical impossibility in any modern economy.

The core result is simple:

$$S(t) < D_c(t) + D_i(t) \quad \text{for all } t > t^*$$

Gold mine production is physically bounded by a logistic ceiling of approximately 3,800 to 4,500 tonnes per year. Global M2 grows exponentially at 6.8% per year with no physical constraint. Industrial gold demand is accelerating with AI adoption. The exponential always overtakes the logistic. This is not debatable — it is a mathematical property of the two function types.

Five systematic robustness challenges fail to overturn this conclusion. The crossover occurs before 2100 even when M2 growth is suppressed to 1%, industrial demand is reduced to 10% of baseline, recycling is doubled, production is scaled to 5×

current levels, or reserve requirements are reduced to near zero.

The four allocation scenarios confirm that there is no way to distribute scarce gold that avoids welfare collapse. Full allocation to the central bank staves off monetary instability at the cost of technological stagnation. Full allocation to industry achieves technological progress at the cost of currency chaos. The compromise distributes both failures simultaneously, amplified by nonlinear coupling in the technology equation.

The conclusion is not that the gold standard is bad policy. It is that the gold standard is geometrically incompatible with an exponentially growing economy. The earth does not produce enough gold each year to simultaneously back an expanding currency and supply an AI-driven industrial sector. No amount of monetary discipline, recycling investment, or geological optimism changes this.

[title=Final Core Conclusion] **The gold standard fails not because governments mismanage it.**

It fails because $S(t) < D_c(t) + D_i(t)$.

Not a policy failure. A mathematical impossibility.

Data and Code Availability

All simulation code is available as `gold_simulation_complete.py`. The complete pipeline runs in a single execution and produces all seven output figures from the raw historical data arrays encoded directly from published sources. No external API dependencies — all data is embedded as verified historical series with source citations.

Data sources: World Gold Council, *Gold Demand Trends Full Year 2024* (February 2025). USGS, *Mineral Commodity Summaries 2025* (January 2025). CEIC/Econovis, *Global Broad Money Supply* (Q2 2024).

About

addn. builds formal models of reality from first principles. This paper is part of the *Signal or Noise Research* series — rigorous quantitative analysis of the questions that matter, written for readers without prior technical background.

The central question across all work in this series is always the same question in different notation: *signal, or noise?*